Response to the additional review

We thank the reviewer for clarifying the design and interpretation issues. The revision centers on candidate composition and tree-topology access in public HUPD ranking. It separates new conditional controls from the original five repeated refits and excludes confidential-data evidence. We address each major, minor and disclosure point below.

B1. Search boundaries and pair-training coverage

The original omitted-score LambdaMART selected the upper corner, depth 5 with 200 trees, in all 45 fold/refit/profile fits. The informed version selected depth 5/200 in 34 and depth 3/200 in 11. We now disclose those frequencies and preserve the original comparison as a specified historical search, without treating its gap as a capacity-independent result.

The extension crosses depths 3/5/7/9 with 100/200/400/800 trees for both information sets. It uses seed 101 on all three uniform outer folds, with 512/1,024/2,048/all eligible training-query budgets. The first 512 lists and 431/433 features are checked against the saved originals before fitting. Larger budgets extend the same training-query permutation. All validation and test queries remain fixed, and validation alone selects every configuration. Prefix prediction is checked against an independently fitted shorter model. These are capacity/coverage sensitivities on a saved representation, not five additional independent fitting runs. Full grids, timing checkpoints and selected configurations accompany the outputs.

All 24 searches and 384 validation configurations are complete. Omitted/informed means are .565361/.624375 at 512 queries, .590275/.639254 at 1,024, .610046/.649030 at 2,048 and .649674/.673369 using all eligible queries, which are 9,993/9,983/9,966 by fold. Their informed-minus-omitted differences are .059014 [.056601, .061481], .048978 [.046712, .051292], .038984 [.036797, .041213] and .023694 [.021879, .025498]. The intervals are component-only and condition on the selected seed-101 models. Independent metric replay has maximum error 1.11e-16.

The omitted learner selects depth 9 in all twelve expanded fits and 9/800 in eight. At the all-query budget, omitted selects 9/800 in every fold, and informed selects 9/800 twice and 7/800 once. Thus coverage and expanded search narrow the observed difference, while saturation remains unestablished. The all-query grade-3 versus grade-2 difference is .001907 [−.002397, .006097], and grade-3 versus grade-1 is −.004537 [−.007443, −.001635], so the aggregate improvement does not hold at every boundary. The main learning figure now presents the four expanded budgets, and the historical 128/256/512-query figure is retained separately in the supplement.

Locations: Results, “Tree topology and finite-budget feature access,” and Supplement, “Expanded learner capacity and training coverage.”

B2. Tree topology is additional side information

We agree that omitted scores do not imply absent hierarchy. The text no longer says that appended f,h add no underlying information. They combine supplied membership estimates with the IPC parent mapping and therefore add the topology as a fixed aggregation. The original comparison is described by its actual feature sets, pair supervision and fitting budget.

Two new 512-query uniform controls use the same expanded LambdaMART grid. The first appends linear sums of predicted memberships at the class and section levels, with query/candidate vectors, products and absolute differences. The second appends a score computed after a single fixed size-preserving shuffle of the subclass-to-path assignment. The nested ancestor incidence matrices are shuffled together, preserving every class/section size. Flat overlap is permutation invariant and is checked numerically. All four feature conditions are complete at the same 512-query budget and 16-configuration search. Their uniform means are .565361 for score omission, .624375 for true f,h, .583869 for shuffled scores and .637208 for linear ancestors. True minus

shuffled is +.040506 [.038410, .042569], and linear minus true is +.012833 [.010578, .015156]. The linear control supplies 120 additional features and is not coordinate-capacity matched to two appended scores. It also lowers grade-3 versus grade-2 accuracy to .657448 from .706246. We therefore interpret topology access and representation jointly, without claiming one compressed summary is necessary or superior on every boundary. The shuffled result concerns one fixed random topology.

MLP is presented as a supplementary representation diagnostic. Its pointwise squared normalized-grade loss, training-fitted StandardScaler and random pair-level early-stopping holdout are stated. This holdout is not query-grouped and is acknowledged as a limitation. Outer validation and test queries remain separate. The main learner interpretation concerns LambdaMART's ranking-loss comparison.

Locations: “Paired feature controls,” Results topology subsection, Discussion, and Supplement topology controls.

B3. Gate tuning, objective mismatch and saturation

The historical gates used fixed training priors, regularization and training mixing strength, while BestSimple received a validation search. Their historical differences are now explicitly conditional on that setup. The fitted-intercept result is brought into the main interpretation: a feature-dependent gate can beat a loss-matched intercept yet remain below a validation-grid constant. Thus the observed difference cannot be assigned entirely to feature dependence.

The new experiment gives every learned objective and BestSimple exactly 144 validation candidates. Expected- and mean-loss gates cross four priors, three KL weights, two fitting strengths and six evaluation strengths, including .75 and 1. Stabilized ListNet additionally searches coefficient L2 and score temperature. The simple reference searches constant retention including the uniform case and similarity weighting over the same six evaluation strengths. All choices use validation nDCG with the declared tie breakers. The scripts save each fit, objective diagnostics, full validation grids, selection and held-out query outcomes. Analytic gradients and the affine mixing shortcut are independently checked. Historical metrics are not overwritten.

All nine fold/construction cells are now complete. On uniform lists, the retuned expected-loss, mean-loss and regularized ListNet gates differ from the matched simple reference by −.000890, −.000903 and −.000351. Their component-only intervals are [−.001220, −.000559], [−.001236, −.000564] and [−.000671, −.000029]. Retrieval intervals span zero. On stratified lists, regularized ListNet improves the simple reference by .001626 [.001206, .002035]. All 648 optimization fits converge. Selected ListNet coefficient norms range from .507 to 1.552, resolving the historical saturation. None of its nine selected strengths is zero or one. The conclusion is now construction-specific and conditional on the stated search, rather than a general rejection of learned gating.

Locations: Results, “Mixing strength, objective and regularization,” and Supplement, “Matched-budget gate search and learned-score mixing.”

B4. Ancestor/leaf reversal, tuning and strata

The task section now explicitly states that the depth-three term of h equals f. Equal depth weighting allocates one third to leaf overlap and two thirds to broader levels. The qualitative trade-off is therefore analytically motivated, while its measured magnitude depends on prediction error, candidate composition and the lexical blend.

All three ancestor-minus-common-alpha-leaf contrasts now appear with refit-plus-component intervals. Uniform is +.0438 [.0414,.0462], retrieval is −.0107 [−.0115,−.0099], and stratified is −.0375 [−.0428,−.0307]. These are pointwise descriptive contrasts. They are not retrospectively inserted into the original two primary test families. The primary families and their sizes remain explicit.

The descriptive uniform advantage is concentrated in the 7,189 lists without grade-3 candidates, difference +.1078. At counts 1 and 2 the differences are +.0176 and +.0112. Counts 3–4, 5–9 and at least 10 favor the leaf

comparator. The section analysis shows negative differences in B and F, −.0383 and −.0514, respectively. These strata are now reported with their denominators. They prevent interpreting an average gain as universal across queries or topics.

We add independently tuned leaf lexical weights and a depth-weighted ancestor score. Validation searches 15 nonnegative weight triples on a .25 simplex, crossed with five lexical weights. The depth-three-only case can recover the leaf overlap. Independent leaf tuning preserves all three directions. Ancestor minus independently tuned leaf is +.044211 [.041801, .046616], −.010384 [−.011573, −.009163] and −.044135 [−.045793, −.042291]. Depth weighting adds .002606 [.002120, .003104] over equal depth on uniform lists. Every one of the fifteen retrieval and fifteen stratified fits selects depth weights (0,0,1), exactly recovering the independently tuned leaf score. The supervised membership-cosine baseline uses the same saved predicted memberships without a pair learner or ancestor aggregation, scoring .553143, .660201 and .764251. These rows make the effect of local IPC supervision explicit.

Composition queries are now sampled randomly without replacement from each full common-feasible intersection, seed 20261031 + fold. The same 1,500 queries and saved stratified-selected pipeline are used at every count. Ancestor-minus-leaf is +.0499 at count 0, +.0082 at 1, −.0158 at 2, −.0404 at 5 and −.0517 at 10. The reversal lies between 1 and 2 under the declared 3:3:1 allocation and fitted lexical weights. The figure distinguishes this predicted-membership finding from the increasing true-membership oracle curve. No universal two-candidate threshold is asserted. We also degrade the saved memberships toward their training-only mean. At prediction weight .25, primary-label accuracy is .433862 and ancestor minus leaf is −.003358 [−.004903, −.001866]. The full saved predictions give accuracy .607076 and difference +.043302 [.040834, .045732]. Zero prediction weight yields constant score terms and identical lexical rankings. This is a calibration/discrimination sensitivity, not a stronger-classifier experiment.

Locations: “Task and Analytical Calibration,” Results ancestor aggregation subsection, and Supplement static-score, membership-quality and stratified-difference analyses.

B5. Positioning, fair supervision and mixing above a learned ranker

The Introduction now gives a classification-assisted patent-landscaping use case. A fixed IPC proxy can organize a supplied candidate set for topic inspection, while examiner relevance and full-corpus recall require different evidence. The contribution is an empirical evaluation of that proxy, rather than a new application model.

We connect the task to pooling reliability, sampled ranking metrics, graded relevance, hierarchical classification evaluation and patent classification. Seven additional references were checked by exact DOI metadata: Zobel (1998), Krichene and Rendle (2020), Silla and Freitas (2011), Kosmopoulos et al. (2015), Voorhees (2000), PatentBERT (2020) and reciprocal-rank fusion (2009). The references are used for their corresponding evaluation concepts, not as evidence for the present numerical findings.

The main score table separates text baselines without local IPC fitting, document-supervised scoring, pair-supervised learners and local mixing. Table 1 now labels its training columns as local fitting. An absolute gap between a frozen encoder and an IPC-supervised scorer is not interpreted as a hierarchy effect. New membership-cosine and linear-ancestor controls expose more directly comparable information sources.

The main story centers composition and topology. Detailed mask losses and gate diagnostics are supplementary. The original gate comparison acts on the ancestor blend, so it does not establish an improvement of the best overall system. The completed simple-mixing check above expanded informed LambdaMART raises uniform nDCG@10 from .624375 to .633432, difference +.009057 [.008068, .010014], with gains in every outer test part. It scores each neighbor with the full learned model for (q,n). Validation selects the simple operation without reading test grades. This result is conditional on seed 101 and the 512-query budget. The different linear-ancestor condition scores .637208, and the all-query informed model scores .673369. Neither is the base of this mixing check. The graph section is consequently scoped to its measured 512-query comparison, without a claim of improvement above the strongest full-coverage scorer.

Locations: Introduction, information and main-score tables, Discussion, and Results learned-score mixing paragraph.

B6. Public artifacts, upstream supervision and reconstruction

The supplied project repository is reachable, but its inspected public tree still contains the earlier pipeline. We now state this distinction directly. The HUPD reconstruction and replay materials are accompanying revision artifacts until deposited. No new DOI, hosted release or completed public deposit is invented. The original HUPD source remains publicly accessible and pinned by revision/hash. A standalone upload package is prepared, while publication of its contents requires an authenticated repository or deposit action.

Future membership refits now set liblinear random_state to the declared run seed. The historical state was unspecified and is preserved, not retroactively made deterministic. The guide distinguishes exact supplied-state score replay from a fresh fit under the corrected randomness setting.

The PatentSBERTa-linked preprint uses 1,492,294 patents from 2013–2017 for similarity and CPC classification, with 663 CPC subclasses. Its adaptation section describes STS supervision and cross-encoder pseudo-labels for initially unlabeled claim pairs, including 1,143 claim sentences and 3,432 selected pairs. Downstream nearest-neighbor CPC prediction is a separate step. This evidence does not establish direct CPC supervision of every released encoder weight. The pinned model card does not identify an item-level training corpus that certifies HUPD exclusion. We state that uncertainty explicitly.

The encoder-independent BM25+TF-IDF candidate pool uses no PatentSBERTa or grade quotas. The same original retrieval-selected models give ancestor .6180 and common-alpha leaf .6309, paired difference −.0129 [−.0138, −.0120] conditional on seed 101. Informed LambdaMART gives .6542 versus omitted .6303. This sensitivity retains the reversal after removing the frozen encoder from candidate construction. It is not a retrained system or independent relevance assessment.

Locations: “Data and Code Availability,” “Representations and baselines,” Supplement source audit, and the accompanying reproduction guide. The public-deposit action remains open.

C1–C9. Statistical, visual and reporting corrections

- C1. The two eligibility tables are replaced by their algebraic rescaling explanation. Uniform positive-IDCG differences multiply by 18,711/18,356, and retrieval non-all-grade-3 differences by 18,711/18,623. Those identities are not described as new robustness evidence. Residual thresholds .7/.8/.95 retain 17,860/18,307/18,681 common query identities. At .7 the uniform ancestor, LambdaMART and expected-gate contrasts are +.0418, −.0685 and −.0025, with the same directions as the full analysis. Binned residual-similarity results and exact denominators accompany the estimates.
- C2. The candidate-inventory figure now shows exact random expectation and grade-3 count histograms. Its caption describes only displayed quantities. The composition figure emphasizes the signed difference, states random query sampling, marks the empirical crossing and explains why the true-member calibration is a different curve.
- C3. The graph-off column and zero-only graph-off figure panel are removed. The zero selections are stated once in prose. The retained selection panel shows mixing strengths.
- C4. Counts use thousands separators, numeric precision is consistent, and sign-flip values at the 50,000-draw Monte Carlo floor are marked at their resolution rather than as precise small tails. Conditional p-values are distinguished from refit-aware intervals. The main nDCG@10 table is referenced. The supplementary nDCG@10 table contains only the additional graph controls and MLP rows, removing the repeated main rows.
- C5. Grade-boundary comparisons use separate points and descriptive refit-SD bars, without lines across categorical boundaries. Omitted/informed learner colors are distinct. Learning-curve axes show the

measured range and fold points. Composition-control curves show changes from the ancestor blend. Graph captions state the conversion from nDCG to percentage points.

- C6. Defensive provenance phrases are removed from the scientific narrative. The source screen is described as a grouping self-consistency check. Ideal DCG, the selected simple comparator and candidate difficulty are defined when introduced. Candidate construction is the underlying object, while fixed lists and selection curves have separate roles.

- C7. Text cosine is denoted t(q,c), leaving the leaf index separate. The descendant set is defined, and the leaf term within h is stated.

- C8. The manuscript uses “repeated refits.” SD describes five whole-run means on overlapping training folds, not a standard error or five independent corpora. New one-seed extensions carry conditional scope only.

- C9. MLP loss/scaling/early stopping, the composition allocation and random sample, full search grids and selected-configuration frequencies are specified. Validation grades select configurations, and test grades only evaluate them.

D1–D7. Data, declarations and submission details

- D1. HUPD's maintainer README declares CC-BY 4.0, with the older card discrepancy retained in the source audit. The PatentSBERTa code repository contains an MIT license. The pinned checkpoint does not declare a weight license, so no blanket weight license is asserted and weights are not redistributed. Public-deposit status is stated accurately.

- D2. The upstream embedding adaptation, downstream CPC classifier and unknown checkpoint overlap are distinguished as described under B6. Table 1 states local training explicitly.

- D3. OpenAI Codex, based on GPT-6, is named in the acknowledgment with the affected abstract and main/supplementary writing, experiment/analysis/figure code and numerical checks. The citation refers to independently fetched official project documentation. The manuscript does not imply that the documentation commit itself is a model version.

- D4. The author confirms that IJCNN 2022 is published and that the confidential-data manuscript is unpublished and not under consideration elsewhere. The conference paper's shared motivation and distinct nHDP/variational methodology are disclosed in the Introduction and cover letter. Current public numerical evidence is separately specified.

- D5. The author supplied NSFC Grants 62471261 and 62161036 and Inner Mongolia Grant 2024JQ165. Haijie Zhang carries the main research and writing work. Weidong Liu contributes data acquisition and cleaning. The author confirms no conflicts of interest. These statements are now included.

- D6. The HUPD archive field cannot be equated with a fully verified legal filing date. Application 13144833 is 20160122 in HUPD, but its matching public patent index gives January 25, 2010. Application 15330968 agrees at January 14, 2016. We report one discrepancy and one agreement, keep the original records unchanged and describe the cohort as selected by HUPD's filing_date field. No temporal holdout is claimed. Source identifiers, indexed publication dates and page hashes are retained.

- D7. We suggest Research Article for this controlled empirical study. The official guidance also lists Negative Result, but the manuscript has positive and negative findings. No new acceptance, DOI assignment or publication action is claimed by preparing revision materials.

Explanatory table notes now appear in prose. Main and supplementary PDFs contain no semicolon punctuation. Displayed excerpts replace source semicolons with disclosed commas. Both supplied biographies and photographs, including Weidong Liu’s, are retained.